



Diagnostic Value of Fecal Calprotectin in Distinguishing Crohn’s Disease from Ulcerative Colitis in Adults

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Introduction

Inflammatory bowel diseases (IBD), including Crohn’s disease (CD) and ulcerative colitis (UC), are chronic and progressive disorders characterized by periods of relapse and remission. They may lead to irreversible damage to the digestive tract and significantly impair patients’ quality of life. The diagnosis of IBD is often challenging in clinical practice because of the absence of pathognomonic symptoms and the considerable overlap with those of irritable bowel syndrome. Although endoscopy remains the reference method for assessing intestinal inflammation, it is invasive. Fecal calprotectin (FC) is a useful non-invasive biomarker for diagnosis, disease monitoring, and prediction of disease flares. This study aimed to compare FC levels according to the type of IBD—CD versus UC—and disease status—active disease versus remission.

Materials and Methods

1. Patients

A prospective, descriptive, and analytical study was conducted at Annaba University Hospital between February 2023 and September 2024. The study included 77 adult patients, with a mean age of 39.7 years, including 53 patients with Crohn’s disease and 24 with ulcerative colitis. No significant association was observed between sex and IBD type.

2. Methods

Fecal calprotectin concentrations were measured using the enzyme-linked immunosorbent assay (ELISA) method.

Stool sample preparation

After thorough homogenization of the stool sample, approximately 100 mg was collected using a Pasteur pipette and transferred into a test tube. The pipette was then broken inside the tube, leaving a small fragment to serve as a stirrer. The extraction solution was subsequently added, and the mixture was vortexed for 30 seconds. It was then homogenized for 30 minutes using the pipette fragment as a stirrer. Next, 1 mL of the mixture was transferred into an Eppendorf tube and centrifuged for 20 minutes at 7,800 rpm. The supernatant could be tested immediately or stored at –20°C for up to three months.

Test principle

For the ELISA-based determination of fecal calprotectin, the sample is diluted and added to wells that have been pre-coated with a specific antibody. This antibody captures the calprotectin present in the sample. Subsequently, a second enzyme-conjugated antibody binds specifically to the captured calprotectin. The enzyme then catalyzes the conversion of a substrate into a colored product. The amount of calprotectin present in the sample is determined by measuring the optical density of the resulting product. This measurement is compared with a previously established calibration curve generated using calibrators containing known concentrations of calprotectin. The calprotectin concentration in the sample can therefore be determined and expressed in mg/kg.

Results

1. Comparison of mean FC levels between CD and UC during the active phase

The mean FC level was higher in patients with UC than in those with CD. However, these results did not demonstrate a statistically significant difference in FC levels between the two groups (Table 1).
Table 1. Comparison of mean FC levels between CD and UC during the active phase

	IBD	N	Mean mg/kg	SD
FC	UC	14	531,11	780,1
	CD	24	249,88	326,24
Student test	P=0,09			

2. Comparison of the mean fecal calprotectin level between active disease and remission in ulcerative colitis

The mean fecal calprotectin level was higher in patients experiencing a flare than in those in remission. However, despite this difference in mean values, the comparison between the two groups did not reveal a statistically significant difference (Table 2).
Table 2. Comparison of the mean fecal calprotectin level between active disease and remission in ulcerative colitis

	Clinical status	N	Mean mg/kg	SD
FC	During a flare-up	14	531,11	751,71
	Outside a flare-up	24	412,6	879,82
Student test	P=0,73			

3. Comparison of mean FC levels between active disease and remission in patients with CD

The mean FC levels were very similar between the two groups: patients with active disease and those in remission. Student’s *t*-test indicated that there was no statistically significant difference in mean FC levels according to the patients’ clinical status (Table 3).
Table 3. Comparison of mean FC levels between active disease and remission in patients with CD

	Clinical status	N	Mean mg/kg	SD
FC	During a flare-up	14	250	320
	Outside a flare-up	24	258	555,53
Student test	P=0,95			

Discussion

Several studies have shown that fecal calprotectin (FC) correlates with inflammatory activity in IBD. A study from Saint-Étienne observed a progressive increase in FC levels before relapse (p = 0.049) . Another study in ulcerative colitis reported significantly higher FC levels in patients with moderate-to-severe disease activity (p = 0.001) . A retrospective analysis from India also found higher FC levels in ulcerative colitis than in Crohn’s disease (p = 0.04) . In our study, mean FC levels were higher in ulcerative colitis than in Crohn’s disease, and higher during flare than during remission; however, these differences were not statistically significant. This lack of significance may be explained by individual variability and confounding factors such as treatment. Thus, FC remains a useful biomarker, but its interpretation should be cautious and context dependent.

Conclusion

In conclusion, although the study showed a trend toward higher fecal calprotectin levels in patients with ulcerative colitis compared with those with Crohn’s disease, as well as during flare compared with remission, these differences were not statistically significant. This lack of significance may be explained by the variability of FC levels and the relatively small sample size. Larger cohort studies are needed to confirm these findings. Nevertheless, fecal calprotectin remains a key marker for assessing intestinal inflammation and is useful for both the diagnosis and monitoring of IBD.

References

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